Md Fayek Mubarrat Jamee

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Research Interests

I develop machine learning and computer vision methods that turn real-world video, images and interaction data into measurable, interpretable outcomes. My work spans video-based human movement analysis, biomedical image analysis and human–AI interaction. I am particularly interested in health and learning applications such as remote monitoring, accessible assessment, and AI tools that support rather than replace human reasoning, and in building systems that are practically usable outside the lab.

Specific interests: computer vision · human movement analysis · signal and time-series analysis · biomedical image analysis · mobile and wearable sensing · human–AI interaction · accessible, resource-efficient AI

Education

Bangladesh University of Professionals Dhaka, Bangladesh
B.Sc. in Information & Communication Engineering *Jan 2015 – Jan 2019*

- CGPA: 3.31 / 4.00 | Thesis: *A Comparative Analysis of Machine Learning Classifiers for Detecting Fake News* (published at DUCSE 2018)

Research Experience

Attune: Timing Cognitive Engagement for Novices Using AI Code Assistants [Live demo] [Video] 2026
Independent research, human–AI interaction

- Built a browser-based research prototype that infers over-reliance and struggle from interaction and typing signals (optional on-device facial landmarks) and adaptively delivers AI-generated code through trace-and-predict, lead-and-reveal or verify-first steps.
- Modelled sycophancy as agreement with the learner's own misconception (stated, or detected in their code or message); added skippable steps, "Why this step?" explanations, confidence probes, pre/post concept tests and a questionnaire, with real Python execution in the browser (Pyodide).
- Ran an informal two-session pilot. Session 1 exposed design flaws, which I fixed. In session 2, a participant with no programming background recognised a sycophantic suggestion as wrong (confidence 1/5) but could not repair it, motivating scaffolded repair over prediction for true novices.
- Built for research use: session export, analysis scripts and automated tests (35-check end-to-end test).

Stress-Recovery Measurement Prototype [Video] 2026
Independent research, mobile and wearable sensing

- Early prototype estimating recovery after a stressor from Apple Watch heart rate and phone-camera facial landmarks (landmark coordinates only; no video stored), ending with guided calming exercises.
- Current sessions use demo data; next steps are recording my own sessions and validating Apple Watch heart rate against a reference sensor, with the longer-term aim of studying people with Parkinson's.

Video-Based 3D Hand Kinematics for Parkinson's Motor Assessment [GitHub] 2025
Independent research

- Built a markerless pipeline (MediaPipe, monocular smartphone video) that extracts 3D hand joint trajectories during drawing and writing, as a low-cost complement to wearable IMUs and optical motion capture.
- Engineered 16 kinematic time-series features (SPARC smoothness, normalised jerk, tremor-band FFT power, wrist range of motion, grip stability).
- Proof of concept with two healthy participants under five simulated conditions: features responded strongly to simulated tremor, and hand-movement smoothness (SPARC) decreased in both participants when they simulated micrographia. Large individual variability motivates personalised calibration.
- In a complementary image-based analysis of a public dataset (204 drawings from 15 PD and 13 healthy participants), person-grouped evaluation gave about 60% accuracy, with image entropy the strongest feature ($p = 0.04$): the finished drawing alone is a modest signal, which motivates analysing the drawing process.

PlateletVision: Automated Giant Platelet Detection for Dengue Monitoring [GitHub] 2024
Independent research

- Built a hybrid pipeline: YOLOv8-Nano cell detection plus Otsu-segmented geometric morphometrics for platelet volume estimation from phone-captured microscopy.
- Developed auto-calibration using red blood cell diameter (≈7.5 µm) as an intrinsic scale, removing the need for a scale bar or calibration hardware.

- 96.56% mAP@50 on the BCCD dataset (907 annotated cells); estimated mean platelet volume of 8.61 fL, within the 7.2–11.7 fL reference range (not clinically validated).

FishMeter: Non-Invasive Freshness Assessment [GitHub] 2024
Independent research

- Designed a K-means clustering and HSV statistical system for contact-free fish freshness grading, tested under variable, uncontrolled lighting; lightweight inference suitable for edge devices.

Selected Projects

relaxicise: Pose-Tracked Movement Breaks [Live app] [GitHub] 2026

- Browser app that turns work breaks into exercise; MediaPipe pose tracking verifies 12 exercises using body-size-normalised joint signals with smoothing and hysteresis-based repetition counting. Runs fully on-device (no video leaves the browser); optional Bluetooth heart-rate monitor; unit-tested.

Messi Spatiotemporal Analysis: Event Data and Player Tracking [GitHub] 2026

- Reproducible analysis of StatsBomb open event data (∼600 matches; per-90 metrics, zone densities) with a YOLO + ByteTrack tracking pipeline, ORB/RANSAC camera-motion compensation, and a tracking-evaluation module (MOTA, ID switches) for SoccerNet ground truth.

Publications and Manuscripts

Published

1. M. F. M. Jamee et al., "Fake News Detection using Different Machine Learning Approaches." *Dhaka University Student Conference on Science and Engineering (DUCSE)*, 2018.

In preparation

1. M. F. M. Jamee, "From Drawing Product to Drawing Process: Combining 2D Sketch Analysis with Video-Based 3D Hand Kinematics for Parkinson's Motor Assessment." 2025.
2. M. F. M. Jamee, "PlateletVision: Calibration-Free Giant Platelet Detection for Dengue Telemedicine." 2025.

Industry Experience

Ernst & Young (EY) Dhaka, Bangladesh
Assistant Manager, Technology Consulting *Oct 2024 – Present*

- Lead development of data-driven enterprise applications, coordinating system design, implementation, testing and deployment across a technical team.
- Developed data-processing and visualisation pipelines that turn complex datasets into decision-support outputs.

Senior Software Engineer *May 2022 – Sep 2024*

- Developed high-performance backend systems (Java Spring Boot) and optimised SQL pipelines for large-scale data; authored API specifications and technical documentation for cross-team standardisation.

Southtech Group Dhaka, Bangladesh
Software Developer *Apr 2019 – May 2022*

- Built core business applications (Java, Spring MVC), translating business requirements into production code.

Technical Skills

Movement & signals	MediaPipe (hands, pose, face), 3D kinematic tracking, SPARC, jerk analysis, FFT/spectral analysis, time-series analysis
Machine learning	PyTorch, scikit-learn, SVM, feature engineering, model evaluation, statistical testing
Computer vision	OpenCV, YOLOv8, ByteTrack, image segmentation, morphometrics, homography/camera-motion estimation
HCI & prototyping	Browser-based study systems, interaction logging, pilot protocols, Pyodide, Web Audio, Web Bluetooth
Programming	Python, TypeScript/JavaScript, Java (Spring Boot), SQL, LaTeX
Tools	Git, Linux, Docker, Jupyter, Matplotlib, Seaborn, Vitest

Teaching and Awards

- **Technical Instructor**, Foundations of C Programming workshops for undergraduate students (logic, algorithms, problem-solving), 2017.
- **Runner-Up**, DIU Android App Contest, Daffodil International University (2017); **Runner-Up**, IUT CSE Fest, Islamic University of Technology (2016).